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## Characterizing the catchment area and identifying scientific priorities with communities: An example from the University of Illinois Cancer Center

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### ABSTRACT

#### Background

The University of Illinois Cancer Center (UICC) has developed scalable approaches to determine catchment priorities with communities, including: (1) defining the Special Geographies Areas (SGA) within the catchment area, wherein UICC has a clinical presence; (2) characterizing disparities; and (3) identifying scientific priorities.

#### Methods

Population-level data (2017–2022) were used to characterize the burden of cancer and its determinants across the UICC SGA and overall catchment area, and the nation. Community Health Needs Assessment (CHNA) data (2023) were used to characterize the types of science prioritized by residents from SGA and other areas.

#### Results

Relative to the nation and catchment area, SGA residents face greater adverse social determinants of health (SDOH: e.g. poverty, food insecurity, pollution) and suffer from a greater cancer burden. Among the 1255 CHNA respondents, SGA and other residents prioritized research on lifestyle factors to prevent cancer (47%), multi-cancer early detection strategies (43%), and novel treatments that would result in minimal side effects (31%). Resident endorsements have guided recent UICC scientific investments.

#### Discussion

Multi-method approaches to population and needs assessment data are powerful resources for cancer centers to co-determine scientific priorities, co-plan scientific investments, and co-develop innovative programs driven by cancer data, all with community stakeholders.

### KEYWORDS

Catchment characterization; community engagement; cancer disparities; strategic planning; priority-setting processes

## Introduction

Cancer centers demonstrate their impact and value, in part, through reducing cancer burden among the populations they engage, serve, and elevate. The National Cancer Institute (NCI), scholars, and cancer center leaders have offered standardized definitions, guidance, and norms [1–4], while recognizing the need for cancer centers to exercise flexibility and adaptiveness to address unique and diverse populations.

Today, one major component of tracking the impact of cancer centers on cancer burden is defining the catchment area wherein populations live, characterizing the

catchment area, and identifying priorities with communities. Guided by the NCI, cancer centers define their catchment areas as geographically defined regions that are home to most of the cancer patients they serve [5,6]. Other major deciding factors for catchment definitions include the boundaries of the cancer center's marketing approach, residences of research participants, and input from community and scientific advisory boards [1,4]. Catchment characterization typically involves continuous multidimensional analysis of representative data on cancer outcomes, cancer risks and behaviors, and social determinants of health (SDOH), guided by supplemental

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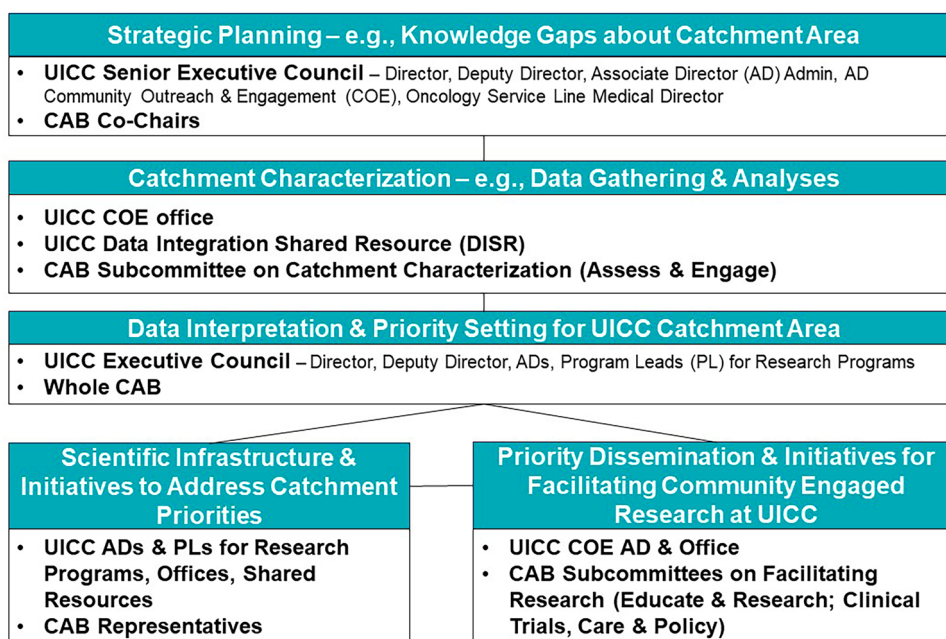
original data collection (e.g. Community Health Needs Assessments (CHNA), focus groups, key informant interviews) [3,7,8]. Scientific priorities for center research, care, and outreach are subsequently co-established and incorporated into center strategic plans, with the guidance of Community Advisory Boards (CABs), town-halls, and other iterative forms of community-academic dialogue [8,9]. Scientific priorities are subsequently disseminated throughout the center and among partners, with the goal of conducting / implementing catchment-relevant research in order to achieve reductions in cancer disparities. The current study highlights the value of these evolving standards and processes for cancer centers to define and characterize their catchment area, with an example from the University of Illinois Cancer Center (UICC).

### Center institutional environment and CAB framework

UICC is an emerging matrix cancer center with a nearly 40-year track record of seminal cancer disparities research [10,11]. UICC maximizes the broadest access to the highest quality of cancer healthcare and research as the ‘cancer hub’ for the University of Illinois Hospital and Health Sciences System (UI Health) and the University of Illinois Chicago (UIC), one of the foremost research-intensive, minority-serving institutions in the Midwest. UICC offers one-of-a-kind access to care and research, in part through its integration with a federally

qualified health center (FQHC). Mile Square Health Center (MSHC) represents one of the few academic FQHCs in the country, with extensive cancer healthcare services (e.g. cancer genetics services, screening services, survivorship services) and a direct, internal route to low / no cost tertiary care (e.g. cancer diagnostic services, cancer treatment services). Beyond institutional resources, UICC fosters a robust referral network with external FQHC and safety net facilities, in part through its institutional charity care and acceptance of all in-state Medicaid plans. Institutional resources have contributed to richly diverse patient and research participant populations at UICC. For example, 79% of UICC interventional treatment accruals were racial/ethnic minorities in 2023, with most patients identifying as Black / African American and Latine.

UICC recognizes that bidirectional community engagement is key to achieving reductions in disparities in its catchment area. The Community Outreach and Engagement (COE) team, patient advocates, and community partners are interwoven throughout UICC operations, including its 16-member CAB. Figure 1 depicts how UICC leverages its multi-dimensional CAB structure for identifying community priorities. First, CAB Co-Chairs and UICC Senior Executive Council engage in strategic planning, including identifying knowledge gaps for the catchment area. Guided by this input, COE facilitates data processes, tools, and collection with the Data Integration Shared Resource (DISR) and a CAB subcommittee dedicated to community needs assessment and evidence-based outreach (‘Assess & Engage’). Results



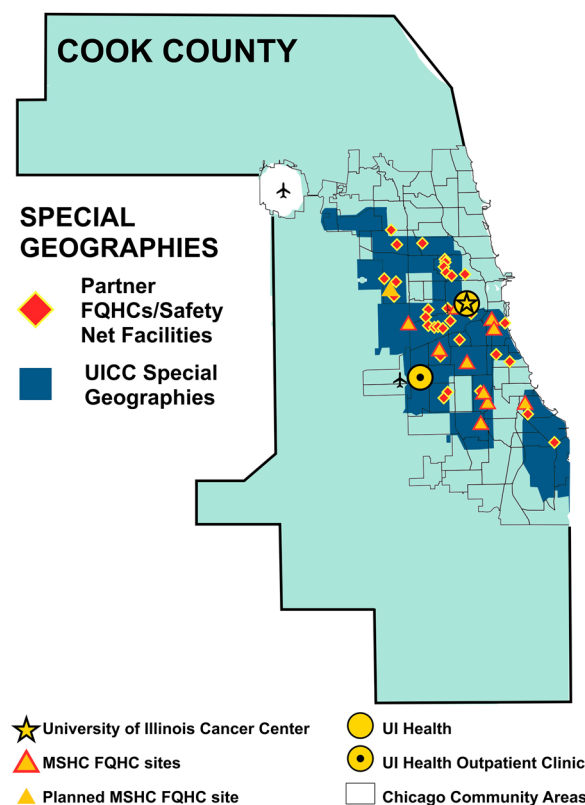
**Figure 1.** Community Advisory Board (CAB) Structure and Framework at the University of Illinois Cancer Center (UICC).

are shared with CAB and UICC leadership for priority setting. COE leadership disseminates data-driven priorities to Associate Directors (AD), Program Leads (PL) for Research Programs, and CAB Representatives, who are dedicated CAB members that work with specific UICC Research Programs, Shared Resources, and offices to actualize catchment-relevant science. COE leadership further facilitates responsive research, guided by two CAB subcommittees focused on the unique needs and capacity for community engaged research in basic, translational, and population sciences (Educate & Research; Clinical Trials, Care & Policy).

### The community case for UICC Special Geographies Areas (SGA)

The UICC Special Geographies Areas (SGA) exemplify how this multidimensional CAB structure has directly improved how UICC characterizes its catchment area and identifies scientific priorities. During 2021–2022, UICC engaged over 600 stakeholders (e.g. UICC CAB, trainees, patient advocates, members, staff) to help develop its 2023–2028 Strategic Plan, entitled *Ascent 2028*. During this process, CAB Co-Chairs advocated to UICC senior leaders the importance of focusing on community areas wherein UICC had a physical clinical presence. CAB Co-Chairs emphasized that these communities would benefit the most from UICC outreach, care, and science, given well-established transportation challenges in the UICC catchment area [12–15]. CAB Co-Chairs advocated that these communities would be most likely to engage UICC for research and care, given their geographic proximity. UICC leadership further recognized that nearby communities may be particularly vulnerable to adverse SDOH, given only medically underserved areas (MUA) are eligible to have FQHCs [16], including MSHC sites. The focus on these communities further aligned with seminal UICC research on health disparities and unequal distribution of MUAs in the catchment area [17]. Through multiple quarterly discussions, UICC Senior Executive Council and CAB Co-Chairs established a definition for the designated UICC SGA and a process for review (e.g. adding new communities) in line with UICC's 5-year strategic planning cycles. The UICC SGA was defined as zip codes wherein there was at least 1 site from UICC's internal healthcare system (UI Health, MSHC FQHC) or 1 FQHC / safety net facility partner who provided active care referrals to UI Health and MSHC. For *Ascent 2028*, 20 zip codes were included in the UICC SGA (Figure 2).

Cancer centers are charged with addressing catchment disparities, as noted above. One example is the importance of defining special populations and identifying special opportunities for health equity in one's



**Figure 2.** UICC Catchment Area, Special Geographies Areas (SGA), Clinical Sites (UIH, MSHC FQHCs), and Clinical Partners (FQHCs, Safety Net Facilities). Notes: UICC = University of Illinois Cancer Center. UIH = UI Health. MSHC = Mile Square Health Center. FQHC = Federally Qualified Health Center.

catchment area. In line with this, UICC conceptualizes its SGA as a 'special population' within its catchment area. UICC thus strives to address the unique disparities faced by SGA and other special populations (e.g. Black residents, Latine residents) within the UICC catchment area with included dedicated resources to match their unique needs, similar to the efforts of other cancer centers focused on racial/ethnic minorities, rural residents, and/or other underrepresented groups [18,19]. Such efforts by UICC and other cancer centers align with the principle of equity, in that catchment residents and subpopulations are given what they need, which is driven by biology, behavior, and SDOH that include environment, clinical and health policies, and care delivery [20]. This differs from efforts for equality, wherein everyone receives the same resources, with limited attention to the rich heterogeneity of needs within catchment communities. Simultaneously, in line with other cancer centers, UICC leverages an intersectionality approach for equitably serving multiple special populations. In line, UICC focuses on addressing disparities among Black and Latine residents in the catchment area, regardless of whether they reside in SGA or not. Similar to cancer centers

that have rural residents as special populations, SGA residents in the catchment area are a priority population for dedicated engagement and services, regardless of racial/ethnic and other identities. UICC thus defines its scope and capabilities in serving its catchment area as a whole, through the lens of equity and intersectionality, wherein its focus is to develop exceptional science and guide care and policy that offers all residents the resources they need, based on their risks.

## Current study

The current study highlights the subsequent processes that UICC COE, DISR, and CAB deployed to characterize catchment cancer burden by geography (SGA, other areas), identify community endorsements of scientific priorities, and incentivize UICC catchment-responsive and community engaged research.

## Materials and methods

### Characterization of the UICC SGA (Table 1)

#### Data sources and measures

Population demographics and adverse SDOH risks were abstracted from publicly available sources, including the US Census American Community Survey, Centers for Disease Control and Prevention (CDC) Behavioral Risk Factor Surveillance System (BRFSS) and PLACES Project, Federal Bureau of Investigation (FBI) Crime Data Explorer, Agency for Toxic Substances and Disease Registry, Environmental Protection Agency (EPA) Environmental Justice Index, EPA Environmental Justice Screening tool, Chicago Crime Data Portal, CDC National Immunization Survey Teen, Health Resources & Services Administration (HRSA) MUA Find tool, and The Gallup Organization [21–29]. We utilized 2018–2022 5-year estimates for age, gender, race/ethnicity, veteran status, disability, education level, unemployment, severe housing problems, food stamp usage, and access to a vehicle, and 2017–2021 5-year estimates for households in poverty and violent crime rate. The most recent 1-year average was used for percent sexual / gender minorities, no health insurance, and air pollution, given the lack of availability of 5-year estimates.

Cancer incidence, late-stage diagnosis, and mortality data were obtained from the Illinois State Cancer Registry, Office of Policy, Planning & Statistics at the Illinois Department of Public Health, the US Cancer Statistics Data Visualizations Tool, and the NCI Surveillance, Epidemiology, and End Results Program (SEER\*Explorer) [30–32]. The current study focuses on UICC priority cancer sites, which are breast, prostate, lung, colorectal, and

liver cancers. Priority cancers were selected by the UICC Senior Executive Council and CAB based on frequencies and disparities in the UICC catchment area as well as prevalence among UICC's cancer cases. The most recent available 5-year estimates for cancer incidence and late-stage diagnosis across zip codes, county, and the US were 2016–2020. The most recent available 5-year estimates for cancer mortality were 2015–2019.

## Analyses

For comparisons in the current study, elevated risks were defined as having a difference of  $\geq 10\%$  and non-overlapping 95% confidence intervals. Estimates were first compared between the UICC SGA and the US, and secondarily between the UICC SGA and the UICC catchment overall.

### Catchment endorsements for scientific priorities

For the current study, the 2023 UICC CHNA was the primary data source for determining resident endorsements for future scientific priorities at UICC (Tables 2–4). The 2023 CHNA was co-designed during Spring 2023 by COE leaders and the CAB Assess & Engage subcommittee, with input from COE Liaisons from the Research Programs, UICC ADs and PLs for the Research Programs, the Clinical Trials Office, and Shared Resources. Examples of CAB contributions to the CHNA tools included data collection on understudied, underserved communities (e.g. new questions to assess veteran status) and the importance of directly engaging communities about scientific priorities, given the research focus of UICC (i.e. outreach, care, and policy must be tied to UICC research).

The UIC Institutional Review Board approved the CHNA data collection efforts as exempt (STUDY2023-0633). Subsequently, the COE team and trusted community partners distributed CHNA surveys in person and electronically throughout the UICC catchment area. During May - September 2023, respondents were recruited during special health events (e.g. health fairs, community townhalls) and UICC COE activities (e.g. community lab mixers, scientific dissemination events). Recruitment strategies and venues reflected strategic efforts to produce an overrepresentation of UICC special populations, including Black, Latine, and SGA residents. Team members initially conducted eligibility screening (i.e. 18 years or older, community member) and obtained verbal consent after sharing standard components (e.g. voluntary nature of evaluation, privacy/confidentiality, benefits and risks of participation). Interested, eligible respondents largely preferred to complete CHNA surveys by themselves, but COE team members were available

**Table 1.** Social Determinants of Health (SDOH), Cancer Risks, and Cancer Outcomes in the US, UICC Catchment Area, and UICC Special Geographies Areas (SGA).

|                                                             | US                     | UICC Catchment         | UICC Special Geographies Areas (SGA) <sup>a</sup> |                          |                          |                           |                          |                          |                           |
|-------------------------------------------------------------|------------------------|------------------------|---------------------------------------------------|--------------------------|--------------------------|---------------------------|--------------------------|--------------------------|---------------------------|
| <b>Demographics</b>                                         |                        |                        |                                                   |                          |                          |                           |                          |                          |                           |
| Population                                                  | 331,097,593            | 5,225,367              | 1,300, 834                                        |                          |                          |                           |                          |                          |                           |
| Age, Median                                                 | 39                     | 38                     | 35                                                |                          |                          |                           |                          |                          |                           |
| Gender (%)                                                  |                        |                        |                                                   |                          |                          |                           |                          |                          |                           |
| Male                                                        | 50                     | 49                     | 49                                                |                          |                          |                           |                          |                          |                           |
| Female                                                      | 50                     | 51                     | 51                                                |                          |                          |                           |                          |                          |                           |
| Race/Ethnicity (%)                                          |                        |                        |                                                   |                          |                          |                           |                          |                          |                           |
| Non-Latine White                                            | 59                     | 41                     | 17                                                |                          |                          |                           |                          |                          |                           |
| Black                                                       | 12                     | 22                     | 31                                                |                          |                          |                           |                          |                          |                           |
| Latine                                                      | 19                     | 26                     | 45                                                |                          |                          |                           |                          |                          |                           |
| Asian / PI                                                  | 6                      | 8                      | 5                                                 |                          |                          |                           |                          |                          |                           |
| Other                                                       | 4                      | 3                      | 2                                                 |                          |                          |                           |                          |                          |                           |
| Sexual / gender minorities (%)                              | 7                      | 6                      | --                                                |                          |                          |                           |                          |                          |                           |
| Veterans (% adult civilian population)                      | 7                      | 4                      | 3                                                 |                          |                          |                           |                          |                          |                           |
| Disability (%)                                              | 13                     | 11                     | 12                                                |                          |                          |                           |                          |                          |                           |
| <b>Social Context</b>                                       |                        |                        |                                                   |                          |                          |                           |                          |                          |                           |
| Families live below FPL <sup>b</sup> (%)                    | 13                     | 13                     | 19                                                |                          |                          |                           |                          |                          |                           |
| High school degree (%)                                      | 89                     | 88                     | 80                                                |                          |                          |                           |                          |                          |                           |
| Unemployed (%)                                              | 5                      | 7                      | 10                                                |                          |                          |                           |                          |                          |                           |
| Violent crime (per 100,000)                                 | 369                    | 539                    | 1070                                              |                          |                          |                           |                          |                          |                           |
| MUA/P <sup>c</sup> Designated Census Tracts (% zips)        | 27                     | 84 <sup>d</sup>        | 100                                               |                          |                          |                           |                          |                          |                           |
| No health insurance (% adults)                              | 11                     | 11                     | 19                                                |                          |                          |                           |                          |                          |                           |
| Severe housing problems (%)                                 | 14                     | 17                     | 21                                                |                          |                          |                           |                          |                          |                           |
| Food stamps (%)                                             | 12                     | 15                     | 25                                                |                          |                          |                           |                          |                          |                           |
| No vehicle available (%)                                    | 8                      | 18                     | 24                                                |                          |                          |                           |                          |                          |                           |
| <b>Physical Context</b>                                     |                        |                        |                                                   |                          |                          |                           |                          |                          |                           |
| Proximity to roads, railways, airports (%ile)               | 33                     | 59                     | 62                                                |                          |                          |                           |                          |                          |                           |
| Traffic intensity (distance weighted vehicles)              | 54                     | 1061                   | 1432                                              |                          |                          |                           |                          |                          |                           |
| Air pollution (fine particulate matter, µg/m <sup>3</sup> ) | 7                      | 10                     | 11                                                |                          |                          |                           |                          |                          |                           |
| <b>Cancer Outcomes</b>                                      | <i>Inc<sup>e</sup></i> | <i>LS%<sup>f</sup></i> | <i>Mort<sup>g</sup></i>                           | <i>Inc<sup>e,h</sup></i> | <i>LS%<sup>f,h</sup></i> | <i>Mort<sup>g,h</sup></i> | <i>Inc<sup>e,h</sup></i> | <i>LS%<sup>f,h</sup></i> | <i>Mort<sup>g,h</sup></i> |
| All cancers                                                 | 442                    | 46                     | 149                                               | 439                      | 51                       | 152                       | 473                      | 50                       | 200                       |
| Breast (F)                                                  | 127                    | 32                     | 20                                                | 131                      | 35                       | 22                        | 129                      | 38                       | 16                        |
| Prostate (M)                                                | 111                    | 23                     | 19                                                | 120                      | 26                       | 22                        | 124                      | h                        | 12                        |
| Lung                                                        | 54                     | 71                     | 35                                                | 52                       | 71                       | 33                        | 61                       | 69                       | 44                        |
| Colorectal                                                  | 37                     | 64                     | 13                                                | 40                       | 65                       | 14                        | 46                       | 63                       | 21                        |
| Liver                                                       | 9                      | n/s                    | 7                                                 | 9                        | n/s                      | 7                         | n/a                      | n/s                      | 12                        |

<sup>a</sup>20 zip codes that represent SGA for 2023-2028 and currently have MSHC and external FQHC/safety net partners. <sup>b</sup>Federal poverty level. <sup>c</sup>HRSA-designated Medically Underserved Areas or Populations (MUA/P), <sup>d</sup>Data limited to the City of Chicago, <sup>e</sup>Incidence = age-adjusted incidence, 2016-2020, <sup>f</sup>LS% = % of cancers diagnosed at regional or distant stage, 2016-2020, <sup>g</sup>Mort = Age-adjusted mortality, 2016-2020, except for SGA (state cancer registry provided 2015-2019 data, without age-stratification due to privacy concerns), <sup>h</sup>Comparison is to US. Data sources: U.S. Census, Behavioral Risk Factor Surveillance System (BRFSS), American Community Survey (ACS), PLACES, National Health Interview Survey (NHIS), Federal Bureau of Investigation (FBI) Crime Data, Environmental Protection Agency (EPA) Mapping. N/A = not available (small cell sizes). N/S = Does not currently have population screening guidelines. **Orange: SGA risk > US for risk; Red: SGA risk > US and overall catchment area for risk; Green: SGA risk < US and overall catchment for risk**

for respondents with limited health literacy, general literacy, and/or writing skills. Surveys were available in English and Spanish. On average, surveys were completed within 10 minutes. Compensation included UICC branded materials, such as high quality water bottles.

### Measures

**Demographics.** Standard close-ended questions were used to collect data on age ('How old are you?'), race/ethnicity ('What is your race [Please select all that apply]?', 'Are you of Hispanic or Latino origin?'), gender ('What gender do you most identify with?'), sexual / gender minority status ('Do you identify as LGBTQ?'), veteran status ('Have you ever served on active duty in the US Armed Forces, Military Reserves, or National Guard?'), disability status ('Do you identify as a person with a disability or differently abled?'), educational

attainment ('What is your highest education?'), insurance status ('What type of health insurance or health care coverage do you currently have? [Please select all that apply]?'), and preferred language ('What is your preferred language?'). Respondents were also asked about direct experiences with cancer ('Have you ever been diagnosed with cancer?', 'Which of the following people in your family have been diagnosed with cancer?'), which were combined into a binary variable (Yes / No). Response categories were reviewed and collapsed, as needed, based on frequency distributions. Final response categories are provided in Table 2. Self-reported residential zip code data were used to classify residents as from the SGA or other areas of the UICC catchment area.

**Scientific Priorities.** Respondents shared their opinions about what the most important topic(s) were for scientists

**Table 2.** Study sample demographics for Community Health Needs Assessment (CHNA) respondents (n = 1255).

|                                        | UICC Catchment Area <sup>a</sup> | UICC Community Health Need Assessment |                                           |                       | p-value <sup>b</sup> |
|----------------------------------------|----------------------------------|---------------------------------------|-------------------------------------------|-----------------------|----------------------|
|                                        |                                  | Overall (n = 1255)                    | Special Geographies Areas (SGA) (n = 612) | Other Areas (n = 643) |                      |
| <b>Age, Median</b>                     | 38                               | 52                                    | 50                                        | 53                    | 0.52                 |
|                                        | %                                | n (%)                                 | n (%)                                     | n (%)                 |                      |
| <b>Race/Ethnicity</b>                  |                                  |                                       |                                           |                       | < .001               |
| Non-Latine White                       | 41%                              | 70 (6%)                               | <b>24 (4%)</b>                            | <b>46 (8%)</b>        |                      |
| Black                                  | 22%                              | 637 (51%)                             | <b>235 (37%)</b>                          | <b>403 (66%)</b>      |                      |
| Latine                                 | 26%                              | 498 (40%)                             | <b>359 (54%)</b>                          | <b>138 (23%)</b>      |                      |
| Asian / PI                             | 8%                               | 13 (1%)                               | <b>5 (1%)</b>                             | <b>8 (1%)</b>         |                      |
| Other                                  | 3%                               | 37 (3%)                               | <b>20 (3%)</b>                            | <b>17 (3%)</b>        |                      |
| <b>Special Geographies Areas (SGA)</b> | 25%                              | 612 (49%)                             | –                                         | –                     | –                    |
| <b>Gender</b>                          |                                  |                                       |                                           |                       | 0.31                 |
| Male                                   | 49%                              | 225 (20%)                             | 120 (19%)                                 | 135 (22%)             |                      |
| Female                                 | 51%                              | 995 (79%)                             | 520 (81%)                                 | 475 (78%)             |                      |
| Gender minorities                      | –                                | 5 (0.4%)                              | 3 (0.5%)                                  | 3 (0.5%)              |                      |
| <b>Sexual Minorities</b>               | 6%                               | 80 (6%)                               | 34 (5%)                                   | 46 (8%)               | 0.11                 |
| <b>Veteran</b>                         | 4%                               | 38 (3%)                               | 17 (3%)                                   | 21 (3%)               | 0.42                 |
| <b>Disability</b>                      | 11%                              | 183 (15%)                             | 83 (13%)                                  | 100 (16%)             | 0.09                 |
| <b>High school degree</b>              | 88%                              | <b>994 (79%)</b>                      | <b>487 (76%)</b>                          | <b>507 (83%)</b>      | <b>0.002</b>         |
| <b>Uninsured</b>                       | 11%                              | <b>253 (20%)</b>                      | <b>154 (24%)</b>                          | <b>99 (16%)</b>       | < .001               |
| <b>Direct cancer experience</b>        | –                                | <b>734 (59%)</b>                      | <b>357 (56%)</b>                          | <b>377 (62%)</b>      | <b>0.03</b>          |
| <b>Preferred language – English</b>    | 65%                              | <b>935 (75%)</b>                      | <b>402 (63%)</b>                          | <b>533 (87%)</b>      | < .001               |

Notes: UICC = University of Illinois Cancer Center. PI = Pacific Islander. Direct cancer experience = personal diagnosis of cancer or family / friend history of cancer.

<sup>a</sup>Data reflect most recent, available representative population data from the American Community Survey (ACS) and Behavioral Risk Factor Surveillance System (BRFSS).

<sup>b</sup>Bivariate comparisons across CHNA respondents; **p < .05 in bold.**

**Table 3.** Scientific priorities by geography, with bivariate analyses (n = 1255).

|                                               | Overall   | UICC Special Geographies Areas (SGA) (n = 612) | Other Areas (n = 643) | p-value      |
|-----------------------------------------------|-----------|------------------------------------------------|-----------------------|--------------|
| <b>Prevention Priorities for Science</b>      |           |                                                |                       |              |
| Lifestyle                                     | 593 (47%) | 301 (47%)                                      | 292 (48%)             | 0.75         |
| Medicine                                      | 489 (39%) | 261 (41%)                                      | 228 (37%)             | 0.23         |
| <b>Early Detection Priorities for Science</b> |           |                                                |                       |              |
| Self / Home-based Screenings                  | 397 (32%) | 205 (32%)                                      | 194 (31%)             | 0.73         |
| Multiple Cancer Early Detection               | 544 (43%) | 277 (43%)                                      | 267 (44%)             | 0.85         |
| Precision-Based Screening                     | 437 (35%) | 224 (35%)                                      | 213 (35%)             | 0.99         |
| <b>Treatment Priorities for Science</b>       |           |                                                |                       |              |
| Self / Home-based Treatment                   | 257 (21%) | <b>153 (24%)</b>                               | <b>104 (17%)</b>      | <b>0.003</b> |
| Fewest Side Effects                           | 390 (31%) | 207 (32%)                                      | 183 (30%)             | 0.38         |
| Precision-Based Treatment                     | 325 (26%) | 170 (26%)                                      | 155 (25%)             | 0.65         |

Notes: UICC = University of Illinois Cancer Center. **p < .05 in bold.**

to study to address cancer in their communities, using multiple close-ended questions developed by the COE and CAB Assess & Engage subcommittee, in response to what was most appealing to the community (CAB stakeholders) and most feasible (UICC stakeholders). For all questions, respondents were provided with an 'Other' option, which allowed them to provide an open-ended response. Preliminary review of these open-ended responses aligned with provided response categories (e.g. 'diet' as Other instead of Lifestyle). Consequently, responses were recoded to categories described below.

For prevention priorities, a categorical variable was created from responses to one question ('What are the most important types of treatments and medicines to

develop for cancer?'). To note, this question focuses on prevention priorities for individuals with and without a history of cancer, in line with decisions and conceptualizations made by community and academic partners. Under this joint conceptualization, strategies to reduce the risk of cancer (initial or subsequent cancers) *and* recurrent risk were offered as response categories. Original response categories included: (1) No / Multiple Priorities, if no priority or both prevention priorities were selected; (2) Lifestyle, if respondents chose 'New lifestyle strategies (e.g. diet, physical activity) to prevent cancer and help people diagnosed with cancer'; and (3) Medicine, if respondents chose 'Medicine for patients to prevent cancer from starting'.

**Table 4.** Scientific priorities by geography with multivariable, multinomial regression models adjusting for age, gender, education, and direct cancer experience (n = 1255).<sup>a</sup>

| PREVENTION                      |                              |            |         |                                 |            |         |                           |            |         |
|---------------------------------|------------------------------|------------|---------|---------------------------------|------------|---------|---------------------------|------------|---------|
|                                 | Lifestyle                    |            |         | Medicine                        |            |         |                           |            |         |
|                                 | OR                           | 95% CI     | p-value | OR                              | 95% CI     | p-value |                           |            |         |
| Special Geographies Areas (SGA) |                              |            |         |                                 |            |         |                           |            |         |
| Yes                             | 0.85                         | 0.67, 1.12 | 0.24    | 0.99                            | 0.71, 1.37 | 0.94    |                           |            |         |
| No                              | REF                          | REF        | REF     | REF                             | REF        | REF     |                           |            |         |
| EARLY DETECTION                 |                              |            |         |                                 |            |         |                           |            |         |
|                                 | Self / Home-based Screenings |            |         | Multiple Cancer Early Detection |            |         | Precision-Based Screening |            |         |
|                                 | OR                           | 95% CI     | p-value | OR                              | 95% CI     | p-value | OR                        | 95% CI     | p-value |
| Special Geographies Areas (SGA) |                              |            |         |                                 |            |         |                           |            |         |
| Yes                             | 1.26                         | 0.89, 1.78 | 0.19    | 1.26                            | 0.93, 1.72 | 0.14    | 1.31                      | 0.94, 1.83 | 0.12    |
| No                              | REF                          | REF        | REF     | REF                             | REF        | REF     | REF                       | REF        | REF     |
| TREATMENT                       |                              |            |         |                                 |            |         |                           |            |         |
|                                 | Self / Home-based Treatment  |            |         | Fewest Side Effects             |            |         | Precision-Based Treatment |            |         |
|                                 | OR                           | 95% CI     | p-value | OR                              | 95% CI     | p-value | OR                        | 95% CI     | p-value |
| Special Geographies Areas (SGA) |                              |            |         |                                 |            |         |                           |            |         |
| Yes                             | 1.46                         | 0.82, 2.59 | 0.20    | 0.72                            | 0.50, 1.03 | 0.07    | 0.75                      | 0.51, 1.13 | 0.17    |
| No                              | REF                          | REF        | REF     | REF                             | REF        | REF     | REF                       | REF        | REF     |

<sup>a</sup>All analyses adjusted for age (continuous), gender (female/non-binary, male), education (less than high school, high school, some college, college degree, grad school), and direct cancer experience (no, yes). Notes: CI = confidence interval, OR = odds ratio, REF = reference group = No/multiple priorities.

For early detection priorities, a categorical variable was created from responses to a single item ('What is the most important strategy that scientists should study to find cancer as early and quickly as possible?'). Original response categories for this question drew from emerging screening technologies of interest to the CAB and UICC Research Programs. Response categories included: (1) No / Multiple Priorities, if no or multiple early detection priorities were selected; (2) Self / Home-based Screenings ('Self-screenings / exams that you can do at home'); (3) Multiple Cancer Early Detection ('Screenings that can find multiple cancers at once'); and (4) Precision-Based Screening ('Screenings tailored to you (e.g. family history, age, environment, biological factors)').

For treatment priorities, a categorical variable was created from responses to one question ('What are the most important types of treatments and medicines to develop for cancer?'). To note, treatment priorities included care and interventions to address experienced cancer diagnoses. Original response categories for this question drew from emerging treatment areas of interest to the CAB and UICC Research Programs. Response categories included: (1) No / Multiple Priorities, if no priority or both treatment priorities were selected; (2) Self / Home-based Treatment, if respondents chose 'Medicine that patients can take at home / can give themselves'; (3) Fewest Side Effects, if respondents chose 'Medicine with the fewest side effects'; and (4) Precision-Based Treatment, if respondents chose 'Medicine tailored to you and your experiences (e.g. family history, environment)'.

## Analyses

For CHNA analyses, missingness was first assessed. Primary models reflect list wise deletion techniques, but similar patterns were found when using multiple imputation by chained equations (20 iterations; data not shown). The demographic characteristics of CHNA respondents were first compared to representative data for the UICC catchment overall (Table 2). Second, descriptive statistics (e.g. chi-square tests, bivariate regressions) were used to examine differences between respondents from the SGA and other areas in the UICC catchment area (Table 2). Finally, scientific priorities were compared by geography (SGA, other areas) using bivariate analyses (Table 3) and multinomial regression models (Table 4). Multinomial regression models incorporated standard and theoretical *a priori* covariates, including age, gender, education, and direct cancer experience. For all multinomial regression models, the odds of each priority were calculated by dividing the probability of that priority by the probability of endorsing no or multiple priorities.

## Incentivization of catchment-relevant UICC research

Interim analyses (summary statistics, bivariate analyses by geography) were conducted and shared with the UICC Senior Executive Council and CAB Co-Chairs every two months (July – September 2023), during deployment of the CHNA, for priority setting. COE leadership

iteratively disseminated priorities to the whole CAB, COE Liaisons from the Research Programs, ADs, and PLs for the Research Programs between August 2023 – March 2024. Guided by CAB stakeholders, the ADs and PLs for Research Programs identified opportunities to catalyze catchment-relevant, community engaged research, including study team formation, identification of external, relevant funding opportunities, and creation of internal funding opportunities for investigator-initiated studies that could be responsive to established priorities.

## Results

### Characterization of UICC SGA

The UICC SGA is home to 1,300,834 residents, who represent approximately 25% of individuals living within the broader UICC catchment area (Table 1). SGA residents are 2.6-fold and 2.4-fold more likely to identify as Black and Latine, respectively, and 71% less likely to identify as non-Latine White, relative to the US. Relative to the overall UICC catchment area, SGA residents are also more likely to identify as Latine. SGA residents are also less likely to identify as veterans relative to the US. Other demographic distributions, including by age, gender, and disability status, are relatively comparable across the SGA and the US.

SGA residents are highly vulnerable to adverse social and structural determinants of health. Socioeconomically, these communities face 46% greater poverty, 10% lower educational attainment, and 2-fold greater unemployment than the US. For healthcare access, 100% of census tracts in the SGA are federally designated as MUAs or MUPs with limited primary care access, relative to 27% of census tracts in the US. Further, residents are 73% more likely to be uninsured relative to the US. Other structural risk factors include 50% greater rates of severe housing problems, 47% greater use of food stamps, 3-fold greater rates of exposure to violent crime, and greater exposure to environmental toxins (e.g. 57% greater exposure to fine particulate matter concentration) than the US. In addition, SGA residents are 3 times more likely to have no available vehicle for transportation. Notably, SGA communities further experience the brunt of multilevel risks relative to the UICC catchment area overall, including 5–99% greater rates of poverty, unemployment, violent crime, severe housing problems, food insecurity, lack of a working vehicle, lack of health insurance, and proximity to sources of environmental toxins (e.g. traffic intensity, fine particulate matter concentration).

Exposure to multidimensional risks likely contributes to the cancer burden in the SGA relative to the US. For incidence, the SGA residents face 12–24% greater rates

of breast, prostate, lung, and colorectal cancer than the US. For late-stage cancer, SGA communities experience 17–19% greater rates of late-stage breast and prostate cancer than the US and catchment overall, while facing 2–3% lower rates of lung and colorectal cancer. For mortality, SGA residents face 26–71% greater rates of liver, lung, and colorectal cancer, while facing 20–37% lower rates of breast and prostate cancer, relative to the US and catchment area overall.

Of note, between 2019 and 2023, 85% of UICC cancer patients resided within the catchment area and 65% resided with the SGA. During this same period, 85% of UICC cancer patients that were accrued to interventional treatment trials lived in the catchment area and 45% lived within the SGA.

### Catchment endorsements for scientific priorities

The initial CHNA sample included 1497 individuals. We excluded individuals who did not live in the UICC catchment area ( $n = 107$ ), individuals with missing / invalid zip codes ( $n = 46$ ), and individuals missing one or more *a priori* selected covariates, including age ( $n = 52$ ), education ( $n = 32$ ), gender ( $n = 9$ ), and/or race ( $n = 1$ ). Relative to the excluded sample ( $n = 242$ ), the analytic sample ( $n = 1255$ ) included respondents who were less likely to be SGA residents ( $p < .001$ ), were older ( $p < .001$ ), were more likely to identify as Latine ( $p < .001$ ), were less likely to be disabled ( $p = .006$ ), were less likely to have been diagnosed or have had a family / friend diagnosed with cancer ( $p = .001$ ), and were less likely to prefer speaking Spanish ( $p < .001$ ).

Table 2 depicts the demographic characteristics of CHNA respondents included in the analytic sample ( $n = 1255$ ). In line with CHNA sampling goals, there was an overrepresentation of UICC special populations, including Black (51% vs. 22%), Latine (40% vs. 26%), and SGA residents (49% vs. 25%) relative to the UICC catchment area. CHNA residents were also more likely to be older (median age: 52 vs. 38 years old), more likely to identify as female (79% vs. 51%), more likely to prefer to speak English (75% vs. 65%), less likely to have completed high school (79% vs. 88%), and more likely to be uninsured (20% vs. 11%) relative to the UICC catchment area. Other demographic distributions, including the proportion of sexual/gender minorities, veterans, and disability status, were comparable across CHNA respondents and the overall catchment area. Among CHNA respondents, SGA residents were more likely to identify as Latine (54% vs. 23%,  $p < .001$ ), less likely to have completed high school (76% vs. 83%,  $p = .002$ ), more likely to be uninsured (24% vs. 16%,  $p < .001$ ), less likely to prefer speaking English (63% vs. 87%,  $p < .001$ ), and

less likely to have direct experience with cancer (56% vs. 62%,  $p = .03$ ) compared to residents of other areas of the catchment.

Tables 3 and 4 depict scientific priorities endorsed by CHNA respondents overall and by geography. Overall, CHNA respondents were most likely to endorse the importance of lifestyle strategies to prevent cancer (47%), endorse multiple cancer early detection strategies (43%), and new treatments with the fewest side effects (31%) across different scientific strategies for prevention, early detection, and treatment. There were largely no significant differences between residents from the SGA and other areas in bivariate analyses nor multinomial regression models adjusted for *a priori* selected covariates (age, gender, education, race/ethnicity, direct cancer experience).

### Incentivization of catchment-relevant UICC research

With support from CAB Co-Chairs, the UICC Senior Executive Council determined the following scientific priorities for UICC: lifestyle strategies for cancer prevention, multiple cancer early detection strategies, and treatments with minimal side effects. COE and PLs for the Research Programs have facilitated several responsive research initiatives which focus on bidirectional community engagement and leverage MSHC FQHC sites. Research has been launched that addresses endorsed scientific priorities while focusing on elevated cancer burdens in the SGA (Table 2; e.g. elevated colorectal cancer incidence, late-stage prostate and breast cancer diagnoses), wherein institutional (i.e. MSHC, UI Health) and external FQHC / safety net partners are physically embedded.

For cancer prevention, UICC has integrated catchment priorities focused on SGA in institutional pilot funding mechanisms, and has catalyzed several active collaborations focused on the SGA. For example, there are active collaborations with community and FQHC leaders to test the impact of multi-sectoral, multi-level strategies on improved food access, improved diet quality, and reduced colorectal cancer risks (e.g. microbial metabolites as intermediate endpoints) in specific SGA communities, which have 67–108% higher food insecurity and 15–24% higher rates of colorectal cancer incidence relative to the catchment area and US (Table 1). For early detection, UICC members are leading multiple FQHC-based clinical trials for different liquid biopsy tests within the SGA. COE has facilitated continuous interactions with FQHC leaders on the team and community partners represented on the FQHC Research Council to discuss the trial design, process, and implications. A

major focus of these trials is proactive engagement of men and people biologically assigned as male at birth, given the disproportionately high rates of late-stage prostate cancer in the SGA (Table 1). Finally, UICC members have begun multiple interventional treatment trials (BRE-09, BRE-08) to test regimens that may minimize or prevent common, major side effects among breast cancer survivors, including severe diarrhea and hair loss. For trials, COE has established relationships with community and FQHC / safety net clinical partners (e.g. study CABs), toward the goal of ensuring continuous, bidirectional feedback on trial implementation. These trials are meant to address the greater likelihood of treatment-related side effects, due to more aggressive treatment from greater rates of late-stage breast cancer diagnoses in the SGA (Table 1).

### Discussion

The current study exemplified the value of multidimensional community engagement, which enabled a more thorough characterization of the catchment area and identification of scientific priorities across different community perspectives. Below, we summarize contributions and new directions for catchment definitions, characterizations, and priority setting for UICC and beyond.

Incorporating CAB leadership in strategic planning directly contributed to identification of a new special population for UICC, based on their understanding of the UICC catchment area and communities. Whereas catchment areas must be defined according to the residence of cancer center patients, cancer case density is generally not distributed evenly throughout the catchment area [33,34]. The current study reinforces community awareness of this nuance and highlights opportunities to engage community partners regarding *which* subpopulations may have greater access to and benefit from cancer center research and care. The current study focused on a new geography-based special population; more scholarship is needed to examine how bidirectional dialogue between communities and cancer centers can contribute to the definition of other special populations, including through intentional, data-driven COE efforts *and* other multilevel determinants (e.g. hospital relationships with payors / acceptance of plans, cancer center staff/member demographic composition).

Altogether, the current study and example of the UICC SGA call for a greater consideration of intersections of place and other structural drivers of inequity. Our study highlights the value of geographic comparisons, in line with cancer centers that serve and characterize the unique burdens of cancer and its determinants in urban and rural communities in their catchment area [35–37].

Simultaneously, it advocates for a shift from traditional methods of characterizing urban catchment areas, which often focus on disparities across demographic characteristics (e.g. race/ethnicity, language, sexual and gender minority status, socioeconomic status) [1,2,4,38,39]. For UICC and other cancer centers, special populations are unevenly distributed throughout the catchment area, due to co-occurring structural forces, including classism, racism, and other forms of oppression. For some cancer centers, including those that serve only urban communities and those that serve urban and rural communities, their physical location may not be easily accessible for historically underserved populations. As such, there has been a need for intentional strategies for COE and others to address transit barriers and avoid unintended systematic exclusion (e.g. field offices, 'lodges' for cancer patients, telehealth research). Community recommendations for the SGA and the final UIC operationalization were driven by where UICC is embedded and how UICC operates in areas that have historically been underserved and under-resourced. SGA residents face elevated levels of structural violence, including poverty, lack of insurance, crime, food insecurity, and pollution relative to the catchment overall and the US. For UICC and cancer centers embedded within the communities that suffer the greatest cancer burden, different COE strategies may need to be employed to avoid systemic exclusion of neighboring residents.

Our work calls for more nuanced characterization of catchment areas, which incorporates the cancer center's locations in understanding the spatial distribution of SDOH and unequal cancer burden, to inform subsequent community engagement. Specifically, there are implications for contributing to future iterations of the P30 CCSG and other standards for cancer centers. These standards often guide cancer centers to consider special populations in their catchment area through considering national trends in cancer disparities, demographic composition (e.g. high proportion of residents from specific historically underserved groups), and cancer disparities in the catchment area. Characterization of catchment disparities is often done at the population level, with a focus on reporting disparities in individual-level, proximal factors contributing to cancer burden (e.g. disparities in physical activity, screening). The example of the UICC SGA highlights the utility of mapping where historically underserved populations reside, which can allow for catchment characterization that identifies well-known neighborhood-level SDOH drivers of disparities (e.g. neighborhood poverty, number and quality of healthcare facilities) in urban *and* rural catchment areas. Our study thus highlights opportunities for a multilevel characterization of the burden that catchment residents

face and enables scientific priorities that can be set at the neighborhood level (e.g. interventions focused on persistently impoverished communities). Our work suggests that such catchment characterization may be helpful for all cancer centers, including those serving only urban populations.

It is important to note that there are potential unintended consequences of identifying the SGA as a special population, including fewer resources for populations that suffer health disparities and do not reside in the SGA. Accordingly, UICC focuses its capabilities throughout the catchment area, for all residents, utilizing intersectionality and equity approaches. For example, UICC's comprehensive financial support (e.g. no / low cost care via FQHC sites, acceptance of all in-state Medicaid plans) is available to all catchment residents and beyond, enabling access and resources for racial/ethnic minorities, veterans, and other populations made vulnerable to health disparities, regardless of SGA residence. UICC further has dedicated, culturally tailored resources across each of its special populations (Black, Latine, SGA), through the lens of offering every resident the care and scientific prioritization they need, based on the risks for cancer that they have. Nonetheless, it should be noted that, similar to other cancer centers, the special populations UICC serves do not include *all* populations that suffer from cancer disparities in the catchment area. Cancer centers may determine special populations based on population size, population composition, and capabilities (e.g. staff composition, partners, etc.), which can unintentionally disservice communities that are widely dispersed or for whom data are not consistently collected (e.g. indigenous populations, veterans, Middle Eastern and North African populations). Such decision-making processes are often done in conjunction with community, which usually *should* incorporate representatives from numerous underserved communities. However, greater transparency and guidelines for such decision-making processes may be helpful for advancing and diversifying the mission and directives for health equity at cancer centers. There is also a need for further conceptualization and scholarship on the longitudinal dynamics and evolution of cancer centers for serving health disparity populations, including how scope expands and capabilities grow for focusing on an increasing number of special populations.

Our complex findings on cancer burden also offer important future venues for research. SGA populations appear to suffer from greater incidence rates, overall and for most priority cancers, relative to the US and overall catchment area. Incidence disparities are an urgent call for transdisciplinary cancer prevention research, especially studies that can be delivered throughout the

SGA, wherein UICC has critical primary care partners (e.g. MSHC FQHC, external FQHCs). Findings regarding disparities in late-stage diagnoses and mortality offer a more complex picture that warrants further research. For example, there are greater late-stage diagnoses of breast and prostate cancer in SGA, but lower mortality for these cancers, relative to the US and catchment area. Conversely, SGA residents faced lower rates of late-stage lung and colorectal cancers, but higher mortality rates, than other areas. Future research is needed to understand these complex results, especially differences in biological drivers (e.g. differential prevalence of aggressive / fast-growing cancers) and SDOH (e.g. access and quality of screening versus treatment care across cancer types; differential impacts of adverse SDOH).

The current study also highlighted the feasibility of a new approach to identifying and setting scientific priorities for catchment areas. Our findings support the feasibility of engaging catchment respondents, via CHNAs, about the type of science that they believe is most important for researchers to pursue. These findings highlight a complementary strategy that can be combined in iterative dialogue with CABs, and used in conjunction with focus groups / listening sessions. For the current study, we leveraged a multi-stakeholder approach [40], integrating the community perspectives of established CAB members and community partners, who offered recommendations based on their historic knowledge of scientific infrastructure for cancer disparities (e.g. member of multiple CABs) and lived experience in cancer-focused outreach and engagement. With UICC partners, CAB members helped to facilitate data collection about scientific priorities that were offered to a wider range of catchment residents. This community-academic approach facilitated generation of questions and response categories that could be understood across different levels of literacy and languages (English, Spanish). Further, the CAB and UICC were able to engage in iterative conversations about the types of response categories or scientific priorities that would be represented, including what might be most appealing to the community, per CAB members, and what might be most feasible for UICC. This iterative process facilitated engagement and CHNA completion with catchment residents, who in turn endorsed priorities as CHNA respondents.

Our CHNA findings were interesting, considering the stark SDOH risks and unique cancer burden that SGA residents face relative to the UICC catchment area overall. SGA and other respondents were largely comparable in their endorsements of scientific priorities in prevention, early detection, and treatment. One possibility may be that perceived scientific priorities may vary more by

individual experiences (e.g. cancer survivor vs not) and community norms than by contextual SDOH risks [40]. For example, across different neighborhood contexts, community values and norms may align more readily with lifestyle strategies for health and well-being [41], given these strategies can be implemented by community leaders and can allow for individual autonomy and ownership over personal health [42–45]. Another possibility may be that scientific priorities address SDOH risks that, while experienced to different degrees by SGA and other residents, may differ in the level to which they are perceived to address SDOH risks (e.g. screenings at home vs. fewer screenings overall). Further multilevel investigation is warranted to explore variation in preferred scientific priorities, which UICC is currently exploring through follow-up qualitative focus groups with respondents and CAB members.

While promising, the current study had numerous limitations that must be acknowledged. Cancer statistics from public use databases lag by 2-to-3 years, and therefore do not represent the burden of cancer in real time. In addition, data suppression rules, while necessary to protect patient confidentiality, resulted in missing data for rarer tumors (e.g. liver cancer) in the SGA. Due to data suppression rules, we could only calculate crude cancer mortality rates in the SGA, as we could not obtain counts stratified by age group. Common limitations further impact the interpretation and caution with which our CHNA findings must be treated, including non-representative sampling and demographic differences across analytic and excluded samples, as well as across SGA and other respondents. These findings limit generalizability and may have influenced documented associations. In line, while our study had a high proportion of Black and Latine residents, there was significantly lower representation of Asian / Pacific Islander, sexual / gender minority, and veteran populations. Given catchment composition, there must be strategic efforts for ensuring overrepresentation in future CHNAs, given they may bear a high burden of UICC priority cancers and deserve tailored outreach and care at UICC (e.g. veterans without VA access). Another major limitation concerns the lack of validated, reliable tools for assessing catchment endorsements for scientific priorities. On the one hand, questions and response categories were developed with community, which is considered a strength; however, singular items were used. Response categories may not have been exhaustive of all possible priorities that communities would have endorsed, although preliminary analysis of open-ended responses for respondents who indicated ‘Other’ aligned with provided options (data not shown). Another limitation is that respondents could choose multiple categories and, in

our analyses, referent categories combined respondents who indicated no priorities with respondents that indicated multiple priorities. Future research should explore forced choice and ranking options, which may facilitate analyses and interpretation. Finally, we note challenges in how survivorship was conceptualized for the current study. Cancer survivorship often is broadly defined to ‘encompass the entire cancer continuum, from the initial diagnosis through the remainder of a cancer patient’s life’ [46]. Our community-academic team made the decision to operationalize priorities, such that there were priorities to address prevention, including reducing the risks of recurrent or new cancers among survivors, and distinguished them from priorities that would address experienced diagnoses. This decision was made, in part, given our goals of obtaining a sample of community residents who would vary in exposure and lived experiences with cancer and who may not necessarily distinguish differences in prevention by cancer history. While broader, there may be different priorities by cancer history – these may be best reported and endorsed by cancer survivors, caregivers, and others with important lived experiences. Future efforts in CHNAs are needed, which allow for complementary approaches to understanding community priorities across the cancer continuum, with the understanding that survivorship often spans multiple stages. Relatedly, the same question was used for both prevention and treatment, in part due to how we conceptualized survivorship. How questions were worded may have influenced the types of responses that were prioritized, given prevention is not necessarily considered in terms of treatments and medicines.

## Conclusion

The current study offered an example regarding the importance of engaging community in defining catchment areas, characterizing burdens, identifying special sub-geographies and populations within the catchment, and setting scientific priorities. Findings suggest the value of community-guided characterization of catchment burden and CHNA on priorities for different scientific methods and approaches. Indeed, these values have directly contributed to several emerging initiatives in our cancer center, which are responsive to catchment priorities and embed community in study design and implementation. Our processes and findings highlight that it is critical to engage with communities using bidirectional and assets-based approaches. These approaches and multidimensional engagements across community leaders and respondents can help to build authentic and trusted relationships, which can ultimately maximize the

ability for UICC and other cancer centers to address health disparities within their catchment areas.

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